FAIR Package Registry: Making Healthcare Package Statements Machine-Readable and Actionable

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Abstract. Healthcare package statements (pakketuitspraken) issued by the National Healthcare Institute in The Netherlands (Zorginstituut Nederland) determine which medical interventions are covered by the Dutch basic health insurance package. We developed a proof-of-concept FAIR Package Registry demonstrating how these statements can be transformed into queryable linked data. The project involved analyzing three breast cancer use cases, conducting 12 stakeholder interviews, creating a semantic model reusing existing ontologies (Cochrane PICO, OBI, STATO, RDF Data Cube), and validating three use cases: automated evidence signalling, real-world data evaluation with The Netherlands Cancer Registry (NCR), and budget impact monitoring. We deployed a FAIR infrastructure including a FAIR Data Station with a SPARQL endpoint, a FAIR Data Point registered in the Health-RI catalog, and persistent identifiers via w3id.org. Stakeholder interviews revealed strong enthusiasm. The proof of concept showed that 100% FAIR compliance is achievable and could serve as a basis for a broader FAIR infrastructure for all data products of the National Healthcare Institute.

Keywords: FAIR data, linked data, healthcare package statements, machine-readable, ontology, learning healthcare system, Netherlands

Background

Healthcare package statements (pakketuitspraken, an example can be found on https://w3id.org/zinl/package-statements/aft) issued by the National Healthcare Institute in The Netherlands (Zorginstituut Nederland) determine which medical interventions are covered by the Dutch basic health insurance package. Currently these statements are published only as PDF documents on the National Healthcare Institute's website. They contain valuable structured information about interventions, indications, clinical evidence, and budget impact analyses. By making these statements available in machine-readable formats, it will both create a metadata-based searchable registry as well as support cyclic package management and real-world evidence evaluation.

Methods

We developed a proof-of-concept FAIR Package Registry to demonstrate how package statements can be transformed into queryable linked data. The project involved:

- analyzing package statement structure through three breast cancer use cases (MammaPrint/Oncotype DX, autologous fat transfer, Trodelvy)
- conducting stakeholder interviews (n=12) with healthcare providers, pharmaceutical industry, and registries
- creating a semantic model reusing existing ontologies (Cochrane PICO, OBI, STATO, RDF Data Cube)
- validating through use cases for automated signalling, real-world data evaluation with The Netherlands Cancer Registry (NCR), and budget impact monitoring.

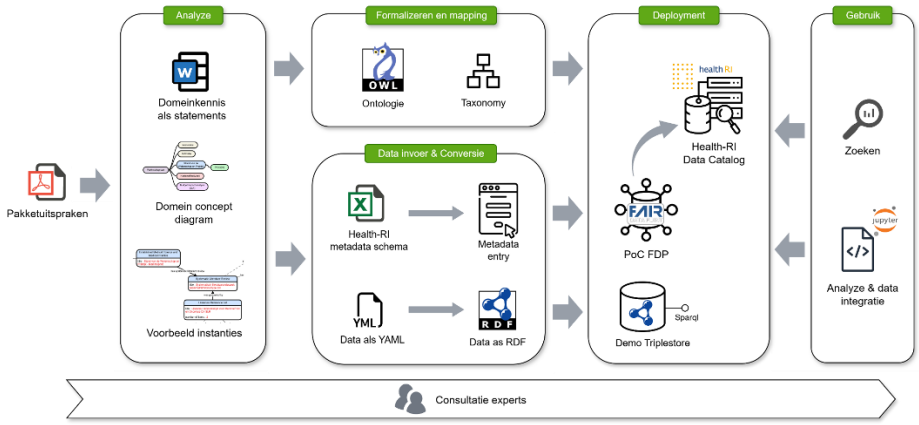

Figure 1; overview of project process (in Dutch)

Results

We created a comprehensive ontology (https://w3id.org/zinl/fpr-o) and taxonomy (https://w3id.org/zinl/fpr-tax) covering package statement metadata, intervention-indication combinations, PICO definitions (Patient/Population (P), Intervention (I), Comparison (C), Outcome (O)), systematic literature reviews, and budget impact scenarios. The model

handles complex variations including multiple intervention-indication combinations, conditional population criteria, and diverse BIA assumption structures.

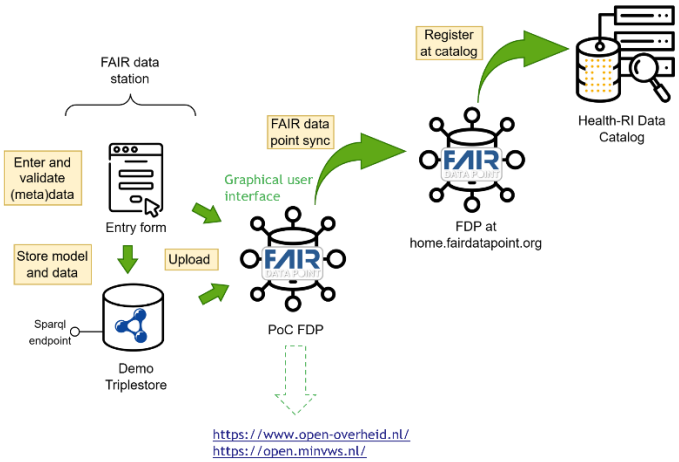

Figure 2: FAIR Infrastructure

We deployed a FAIR infrastructure (https://w3id.org/zinl/fdp) including a FAIR Data Station with SPARQL endpoint, a FAIR Data Point registered in the Health-RI catalog (acceptance environment), and persistent identifiers via w3id.org.

Use case validation demonstrated:

- Automated signalling by storing literature search metadata enabling re-execution to identify new studies and publications.
- Real-world data evaluation through mapping PICO criteria to NCR variables, revealing the importance of early data availability assessment.
- Budget impact monitoring by comparing BIA assumptions with declaration data, showing feasibility when indications map to specific billing codes.

Discussion

Stakeholder interviews revealed strong enthusiasm, and they see the results as an opportunity to data-driven support the learning healthcare system. For the National Healthcare Institute, key opportunities include a better searchable registry, automated evidence signalling and systematic budget impact monitoring. For registries such as the Dutch National comprehensive cancer organisation (IKNL), the organisation that maintains the NCR, value includes systematic evaluation against real-world data and assessment of effectiveness in excluded subpopulations. For healthcare providers and pharma, benefits include easier discovery of coverage decisions.

Key challenges identified include managing structural variations across statements, ensuring data availability for monitoring, and establishing socio-technical foundations for data-driven cyclic package management.

Future work includes AI-assisted extraction from historical statements into YAML files (using a YAML file template and the 3 breast cancer YAML files as training data), alignment with OHDSI Pharmaceutical Research Protocol and OMOP CDM, international HTA collaboration, and integration into workflows of the National Healthcare Institute. The proof of concept showed that it is possible to become 100% FAIR compliant. The PoC could be a basis for a broader FAIR infrastructure for all data products of the National Healthcare Institute and aligns with European Health Data Space requirements and Dutch Open Government Act obligations.

Conclusions

This proof-of-concept demonstrates the technical feasibility of FAIR package statements. While technical infrastructure is proven, realizing learning healthcare systems requires socio-technical alignment with clear governance, established processes, data availability, and stakeholder engagement. Strong stakeholder enthusiasm warrants investment in production-scale implementation.

Acknowledgments

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